

ME 333
MECHANICS OF MATERIALS

Lecture 18: Unsymmetric Bending II &
Tutorial Problems

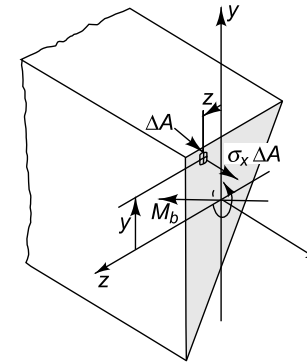
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Sem I, 2025 - 2026

Unsymmetric Bending of Beams

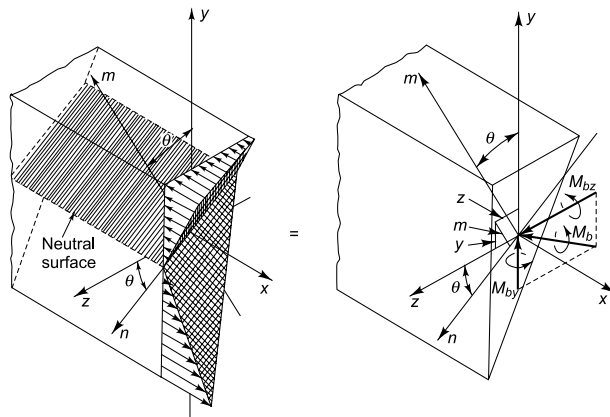
Unsymmetric Bending in Unsymmetric Beams



Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Unsymmetric Bending of Beams

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Unsymmetric Bending of Beams

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Summary

- Respective curvatures in terms of applied moments:

$$\frac{d\alpha}{ds_1} = \frac{d\phi}{ds} \cos \theta = \frac{M_z I_{yy} + M_y I_{yz}}{E(I_{zz} I_{yy} - I_{yz}^2)}$$

$$\frac{d\beta}{ds_2} = \frac{d\phi}{ds} \sin \theta = -\frac{M_z I_{yz} + M_y I_{zz}}{E(I_{zz} I_{yy} - I_{yz}^2)}$$

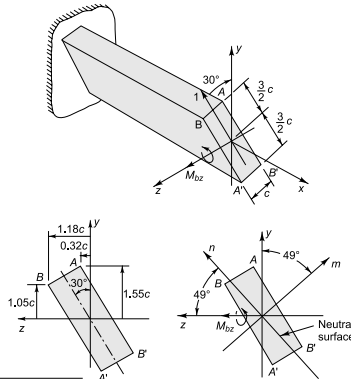
- Stress distribution because of applied moments:

$$\sigma_x = -E \left(\frac{d\alpha}{ds_1} y + \frac{d\beta}{ds_2} z \right) = -E \frac{d\phi}{ds} (y \cos \theta + z \sin \theta)$$

$$= -\frac{(y I_{yy} - z I_{yz}) M_z + (y I_{yz} - z I_{zz}) M_y}{(I_{zz} I_{yy} - I_{yz}^2)}$$

Unsymmetric Bending in Unsymmetric Beams

[Problem 1]: Determine curvature in xy and xz planes. Given: $I_{yy} = 0.75c^4$, $I_{zz} = 1.75c^4$ and $I_{yz} = 0.87c^4$.



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Unsymmetric Bending in Unsymmetric Beams

[Problem 1]: Contd ...

- Using the equation for curvature in xy plane:

$$\frac{d\alpha}{ds_1} = \frac{M_z I_{yy} + M_y I_{yz}}{E(I_{zz} I_{yy} - I_{yz}^2)} = \frac{M_z I_{yy}}{E(I_{zz} I_{yy} - I_{yz}^2)}$$

$$= \frac{0.75c^4}{(1.75c^4)(0.75c^4) - (0.87c^4)^2} \frac{M_z}{E} = 1.36 \frac{M_z}{Ec^4}$$

- Using the equation for curvature in xz plane:

$$\frac{d\beta}{ds_2} = -\frac{M_z I_{yz} + M_y I_{zz}}{E(I_{zz} I_{yy} - I_{yz}^2)} = -\frac{M_z I_{yz}}{E(I_{zz} I_{yy} - I_{yz}^2)}$$

$$= -\frac{0.87c^4}{(1.75c^4)(0.75c^4) - (0.87c^4)^2} \frac{M_z}{E} = -1.58 \frac{M_z}{Ec^4}$$

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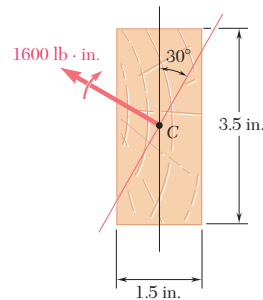
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Unsymmetric Bending

[Example 1]: Determine (a) the maximum stress in the beam and (b) the angle that the neutral surface forms with the horizontal plane.



Ref Base Fig: Mechanics of Materials, Beer & Johnston

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Unsymmetric Bending

[Example 1]: a) the maximum stress in the beam

- Resolve moments about y and z axis :

$$M_z = M \cos(30^\circ) = 1386 \text{ lb.in}$$

$$M_y = M \sin(30^\circ) = 800 \text{ lb.in}$$

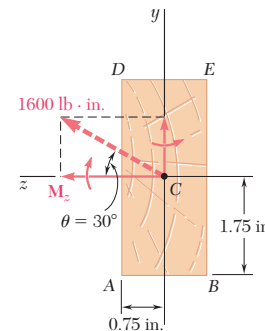
- Determine moment of inertia w.r.to y and z :

$$I_z = 5.359 \text{ in}^4; \quad I_y = 0.9844 \text{ in}^4$$

- Determine maximum stresses due to M_y & M_z :

$$\sigma_x = \frac{M_y z}{I_y} = \frac{800 \times 0.75}{0.9844} = 609.5 \text{ psi}$$

$$\sigma_x = \frac{M_z y}{I_z} = \frac{1386 \times 1.75}{5.359} = 452.6 \text{ psi}$$



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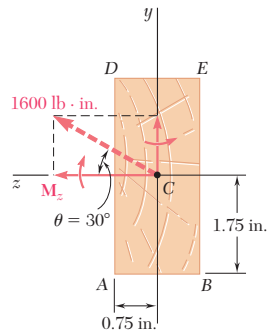
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Unsymmetric Bending

[Example 1]: (a) the maximum stress in the beam



- Total maximum stresses due to M_y & M_z :

$$\sigma_x = 609.5 + 452.6 \text{ psi}$$

$$= 1062.1 \text{ psi}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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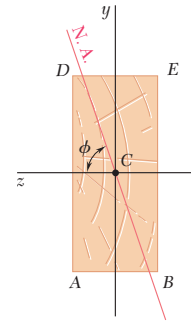
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Unsymmetric Bending

[Example 1]: (b) slope of N.A.



- Slope of the N.A can be determined as :

$$y = \left(\frac{I_z}{I_y} \tan \theta \right) z$$

$$\Rightarrow \tan \phi = \frac{5.359}{0.9844} \tan(30^\circ) = 3.143$$

$$\Rightarrow \phi = 72.4^\circ$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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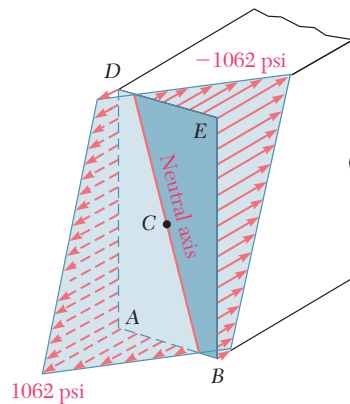
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Unsymmetric Bending

[Example 1]:



Ref Base Fig: Mechanics of Materials, Beer & Johnston

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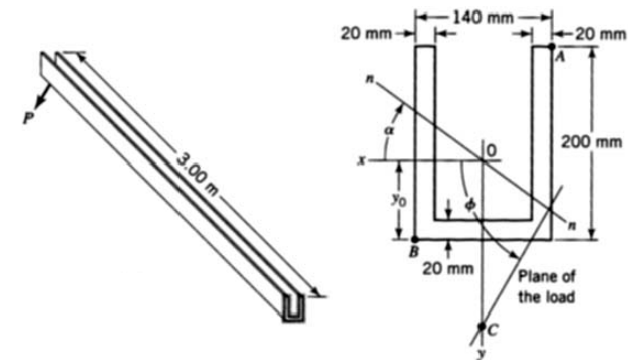
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Unsymmetric Bending

[Example 2]: Locate points of maximum tensile and compressive stresses in the beam and determine the stress magnitudes.



Ref Base Fig: Advanced Mechanics of Materials, Boreisi & Schmidt

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Reading Assignment

Chapter 7 : Crandall et. al.
Chapters 7 & 8 : Boresi & Schmidt